Continuous TFF Operations

Development of a Next Generation TFF system

John Moomaw, Senior Principal Chemist, BR&D
Cyrus Fisher, Senior Advisor, ETC

Outline

- Timeline
- Commercially Available cTFF
- Counter-Current Buffer Exchange (CCBE)
  - Version 1.0 (Manual)
  - Version 2.0 (Manual)
  - Version 3.0 (Manual)
  - Version 4.0 (Automated, Full cTFF)
- Single Pass Filter Screening
  - Configuration optimization
- POC – mAb-R
- Process Development Strategy
- Skid Design & Fabrication

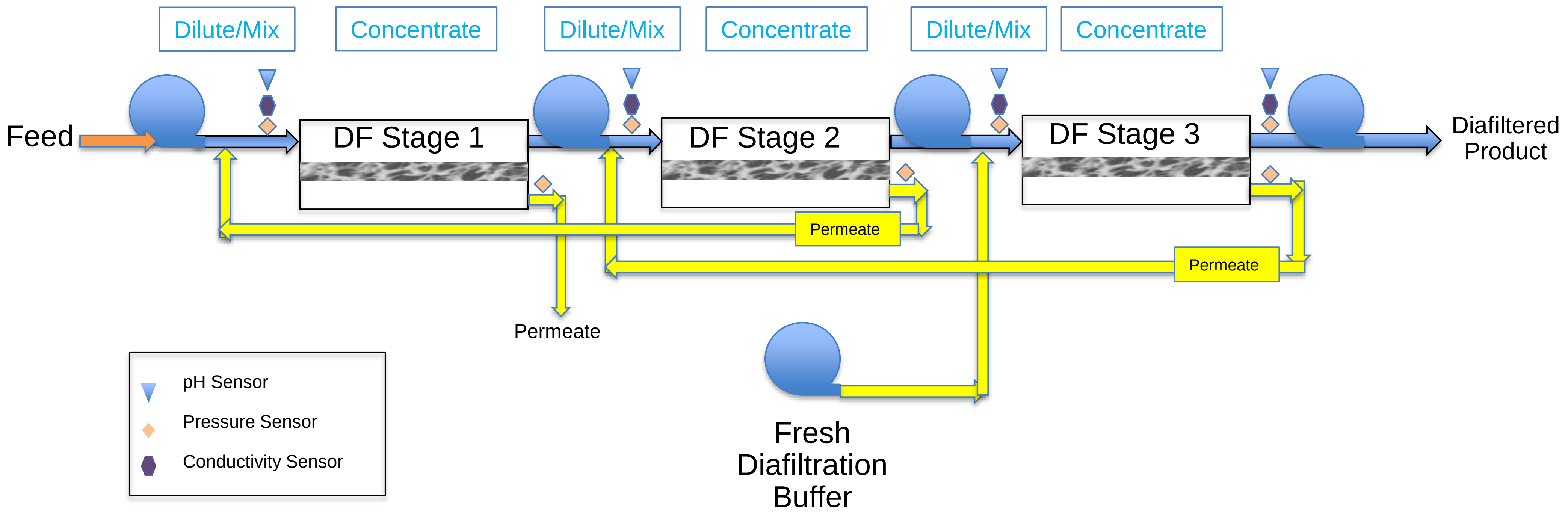
Timeline

- 2015 – Initial Single Pass TFF (SP-TFF) studies
  - Pall & Millipore systems
- 2018 – Single pass buffer exchange testing
  - Pall – In-Line DiaFiltration (ILDF)
    - High buffer utilization
  - Millipore – Alternating 2 tank batch DF
    - Large footprint
  - Zydney – Counter-current Buffer Exchange (Version 1.0)
    - Demonstrated its potential
- 2019 – Counter-current Buffer Exchange (Version 2.0)
  - Pressure/flow testing, improvements to mixing & control
- 2020 – Counter-current Buffer Exchange (Version 3.0)
  - New process control strategy developed and tested
- 2021 – Full cTFF system design & development (Version 4.0)
  - Continuing process control strategy development and testing
  - Developing approach for cTFF filter and operating parameter determination
  - Designing new skid-based automated fully integrated cTFF system
- 2022 – Prototype cTFF Skid Build
- 2023 – Prototype skid testing

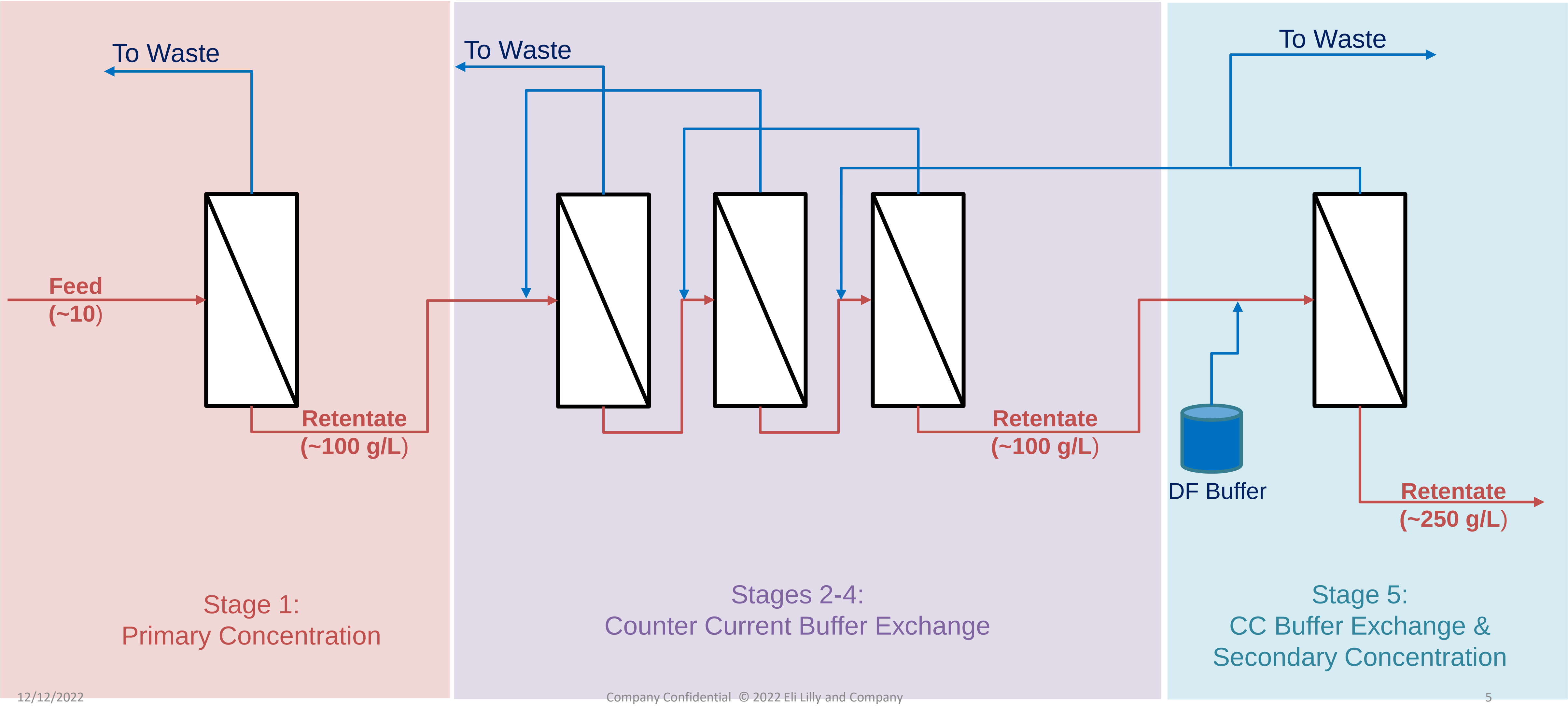
Goal

Have a system ready when needed

Counter-Current Buffer Exchange (CCBE)



Integrated cTFF Design Vision



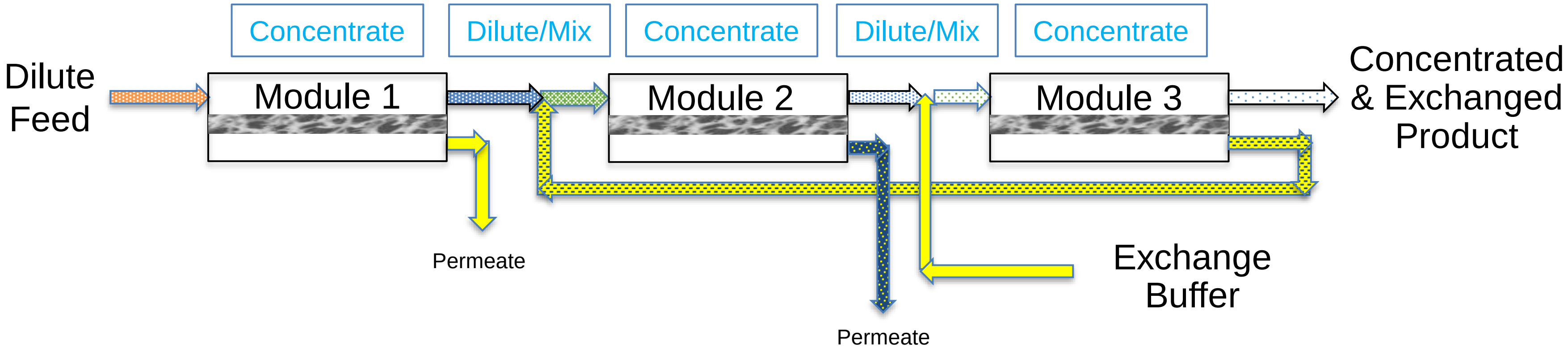
CCBE Lab Version 1.0

- Based on work by Andrew Zydney (Penn State)
  - Biotechnology and Bioengineering. 2018; 115: 139–144.
- Uses spTFF modules and a series of dilution and concentration steps provide buffer exchange.
- Flow ratios (Feed to buffer) control the level of impurity removal.
- Concentration does not change, the matrix does.
- Complex setup, manually controlled.
  - 6 pumps (initially)
  - Used 3 650cm^2 Pall ILC modules (30kDa, RC, smallest option).
  - Difficult to control at high flow ratio (≥10:1) due to low volumetric flow rates.

CCBE Lab Version 1.0 Results

Module 1 Feed: mAb CEX Eluate (20mM Acetate, ~200mM NaCl, pH 5)
Exchange Buffer: 20mM Acetate, pH 5

	Feed to Module 1		Module 1 Retentate		Module 2 Retentate		Module 3 Retentate	
Flow Ratio	Content (g/L)	Osmo (mOsm)	Content (g/L)	Osmo (mOsm)	Content (g/L)	Osmo (mOsm)	Content (g/L)	Osmo (mOsm)
1:1	5.97	254	12.87	258	10.98	164	15.98	99
1:5			29.15	264	23.68	87	35.91	49
1:10			45.63	268	41.39	70	49.15	41
1:20			46.78	273	70.46	66	52.74	41



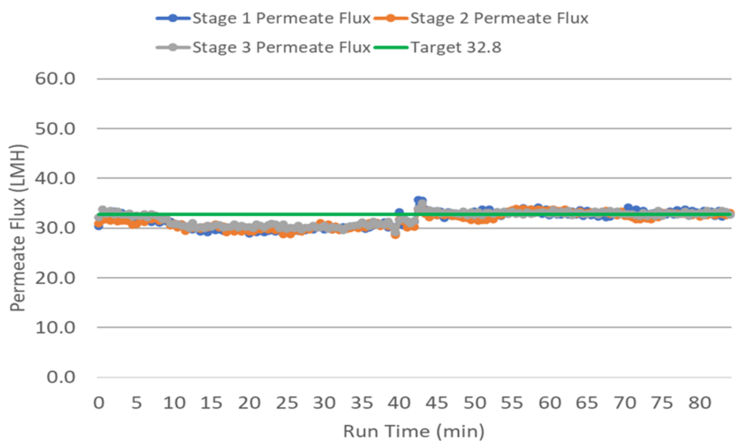
Small scale, low flow rates, manual controls but it appears to work.

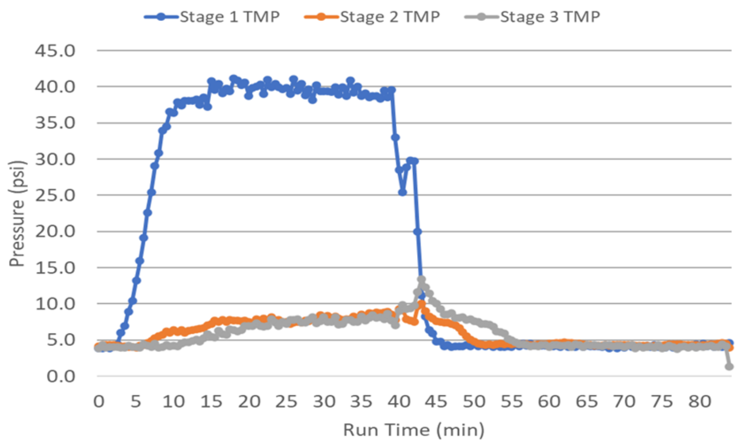
CCBE Lab Version 2.0

- Based on 2nd Zydney publication.
  - Biotechnology Progress. 2019;35:e2810.
- More efficient buffer exchange
- Improved control strategy
  - 5 pumps, flow meters, and needle valves
- Using SPTFF modules (not ILC).
  - More area (1300cm2)
    - Higher volumetric flow rates.
  - 3 modules for buffer exchange
    - No concentration, high exchange factors (>7DV batch equivalent)
  - No internal flow restrictor.
    - Better pressure/flow control.

CCBE Lab Version 2.0 Results

- Demonstrated steady state flow control





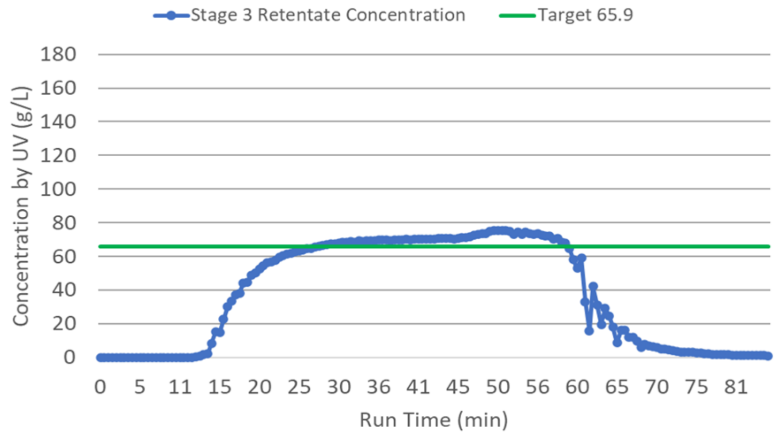
Notes

Feed Conc: 65.9 g/L

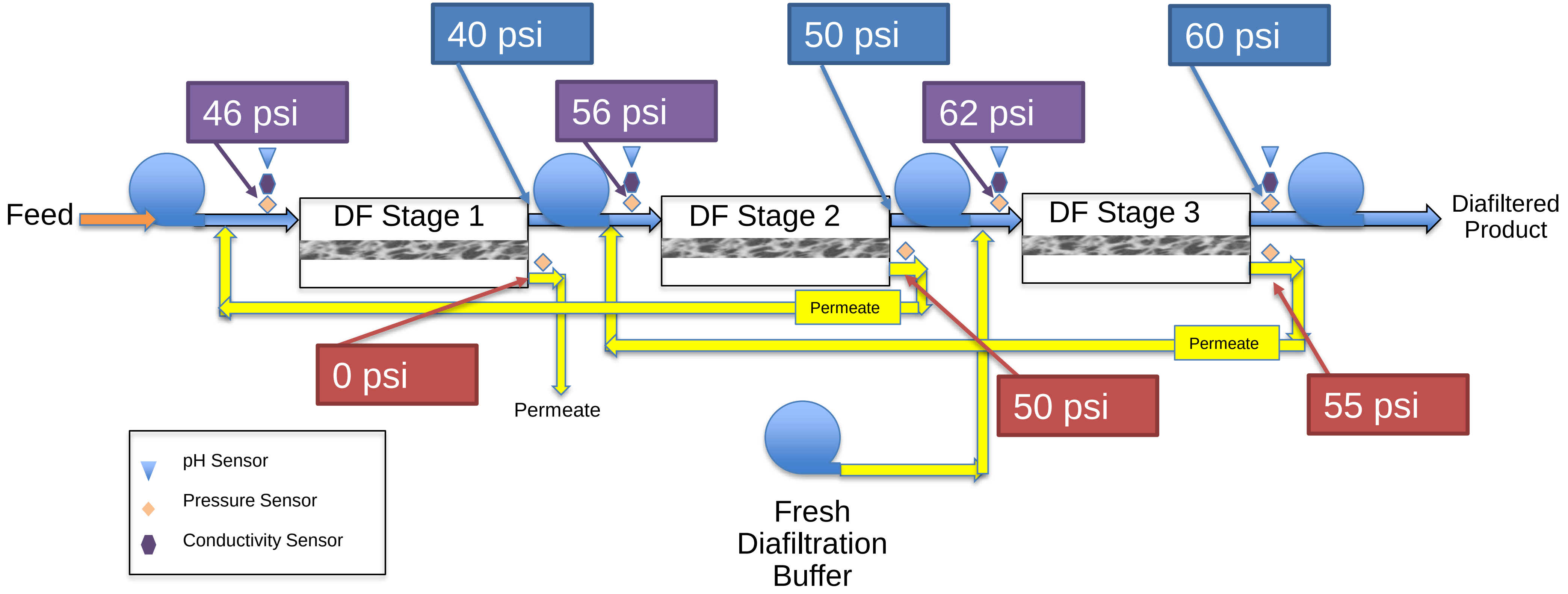
0 min - Begin mAb flow

15-40 min - Adjustments to buffer supply pump to attempt to achieve permeate flux target

40 min - Buffer flush



CCBE Lab Version 2.0 Results (Pressure)



Stage 1
TMP ~43 psi

Stage 2
TMP ~3 psi

Stage 3
TMP ~6 psi

CCBE Lab Version 2.0 Conclusions

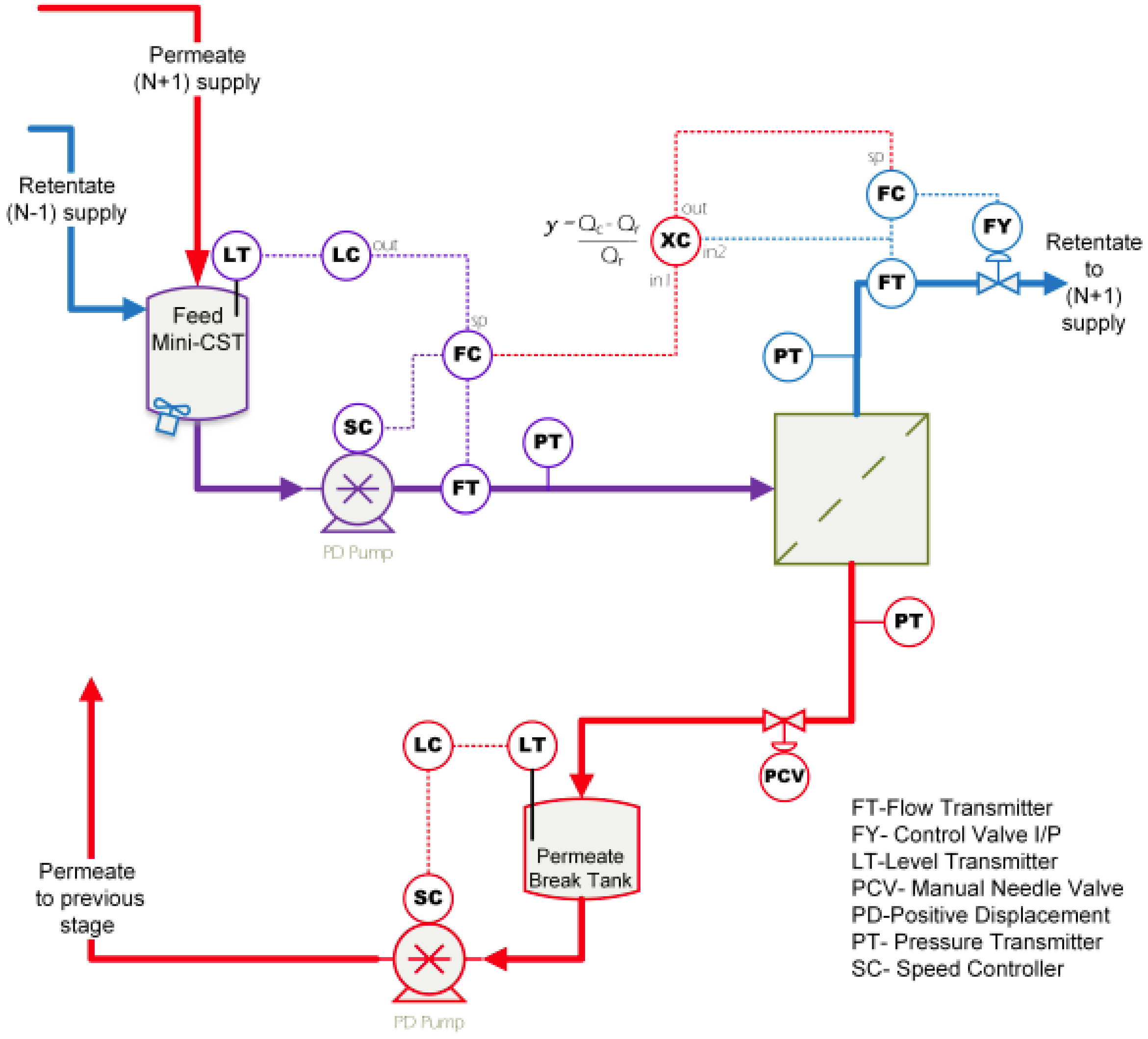
- Demonstrated steady state operation
  - Complex tightly coupled controls required
- Pressure limitations
  - TMP limited after the first stage
  - More stages results in increase pressures (pressure accumulation)
- Could not reach NGB targets for flow ratio or concentration
- Inline mixing challenges

CCBE Lab Version 3.0

Goals

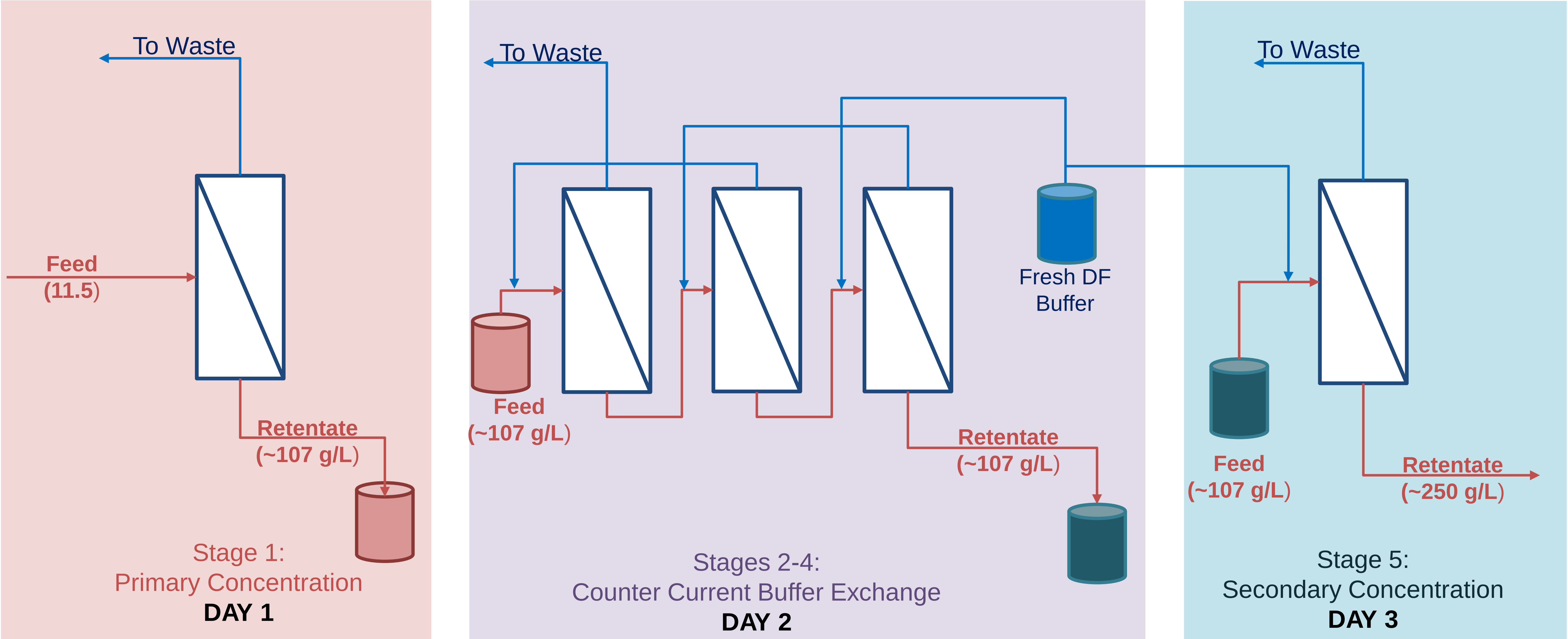
- Demonstrate operation at increased concentration targets (100+ g/L)
- Steady-state for multiple hours (flow, pressure, concentration, etc)
- Still a <u>manual</u> system…
  - If successful, build an automated prototype

The cTFF Module



- A module is adaptable for SP-TFF or CCBE-TFF
- A cTFF system can be build from modules in series
- Mini-CSTs (break tanks)
  - Decouples control between modules
  - Eliminates TMP restrictions and pressure accumulation
- Controls
  - Primary: Level & flow ratios
  - Secondary: Feed flow
  - No direct pressure control

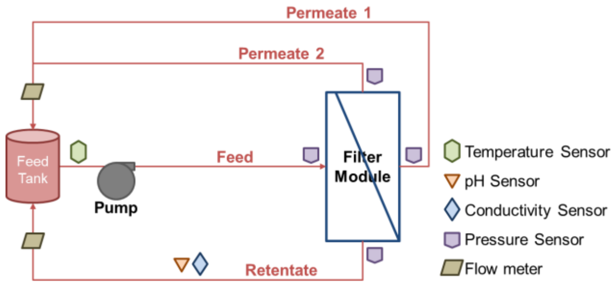
Process in Execution - 3 steps



Selecting Operating Parameters

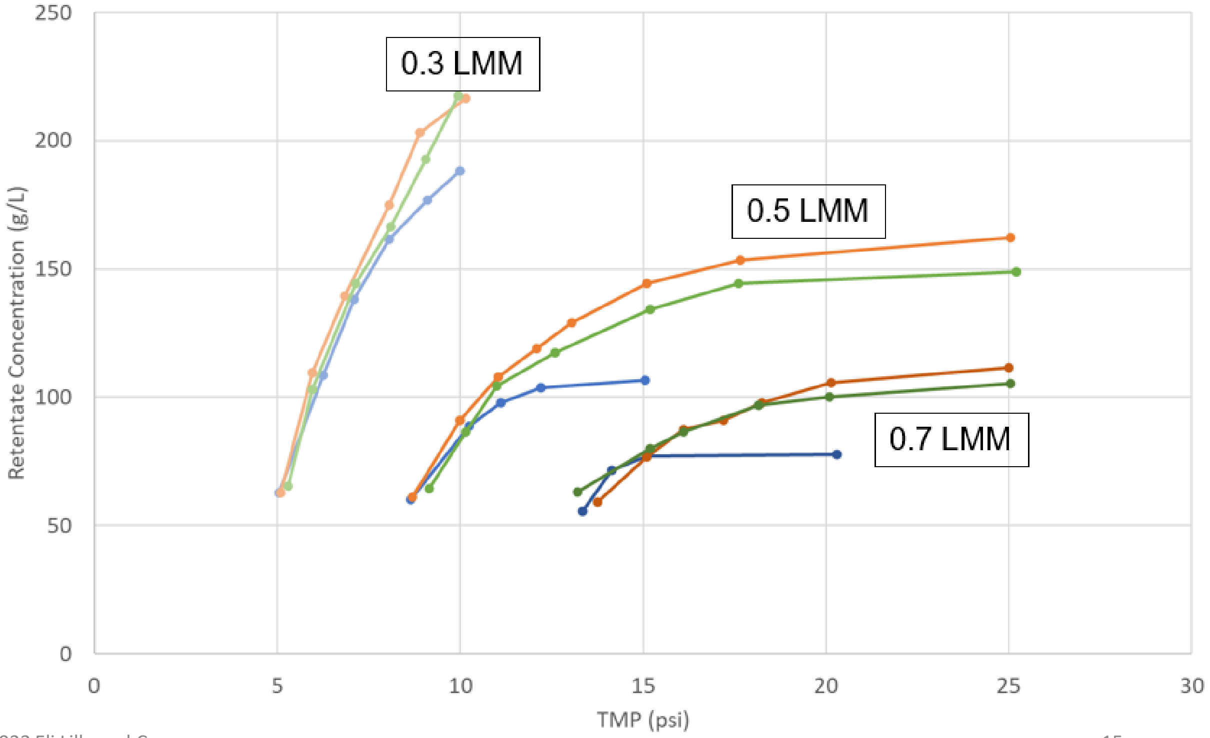
SP-TFF Flux-Excursion Studies

- Use to optimize cTFF operating parameters
  - Crossflow flux, concentration, filter type, filter area, and filter area configuration
  - All stages need not be the same



Example

- Assume target concentrations for feed of 15 g/L, buffer exchange at 100 g/L, and final at 200 g/L
- Primary concentration (blue)
  - Crossflow 0.3 LMM, Expected TMP 6 psi
- Buffer exchange (green)
  - Crossflow 0.6 LMM, Expected TMP 12 psi
- Secondary concentration (orange)
  - Crossflow 0.3 LMM, Expected TMP 10 psi



Primary Concentration

Expected Results

- Primary concentration
  - Crossflow 0.3 LMM (18 LMH)
  - Expected TMP 6 psi

Notes

Feed Conc: 11.5 g/L

Retentate Conc: 100 g/L

0 min – Began recording data

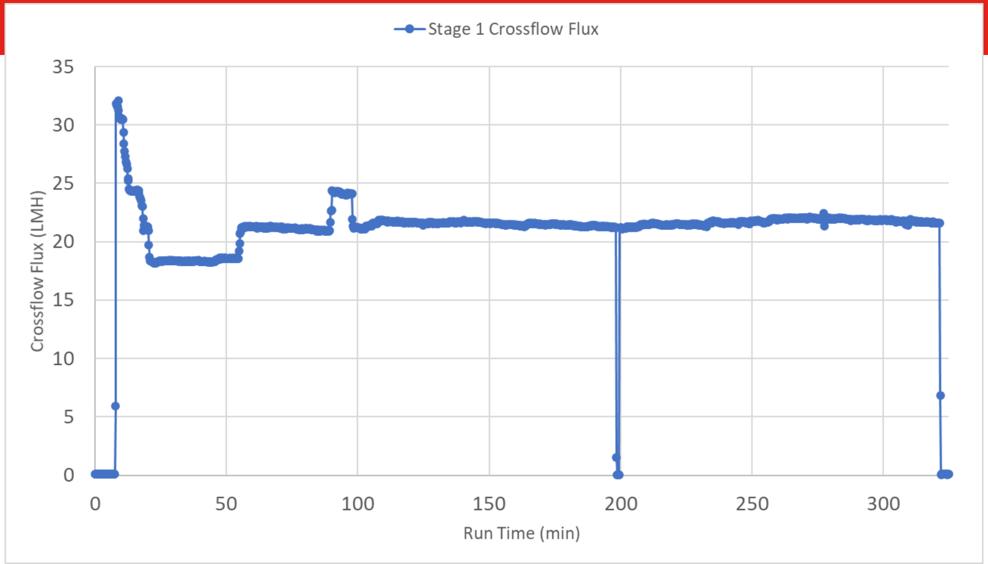
8 min – Started feeding mAb

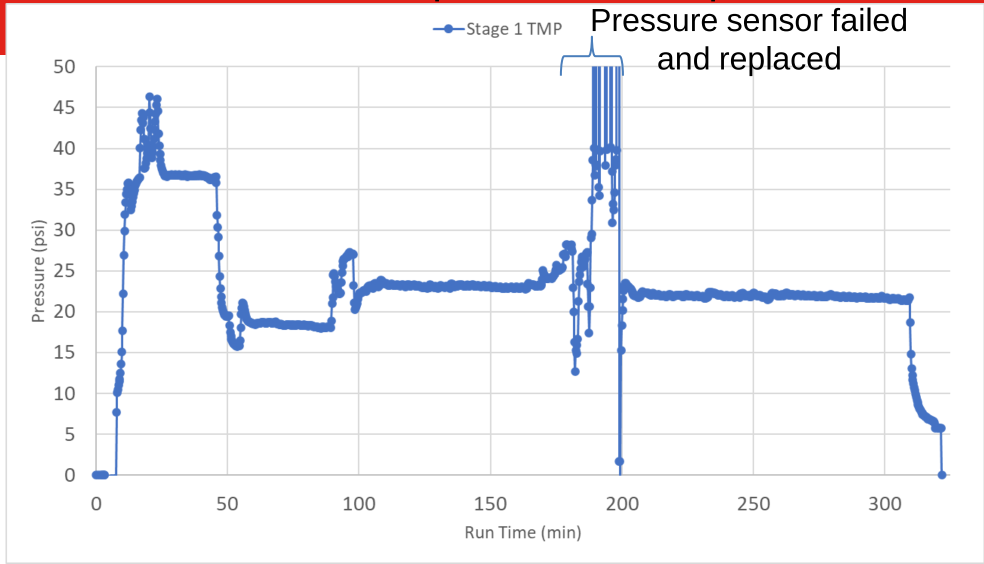
8-100 min – Varying feed flow rates to evaluate system response

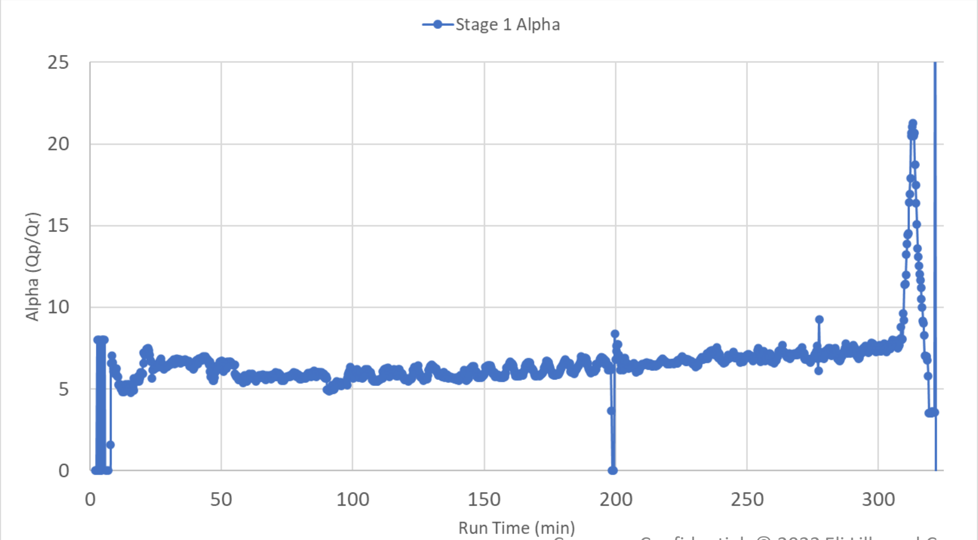
100-302 min – Stable operating period

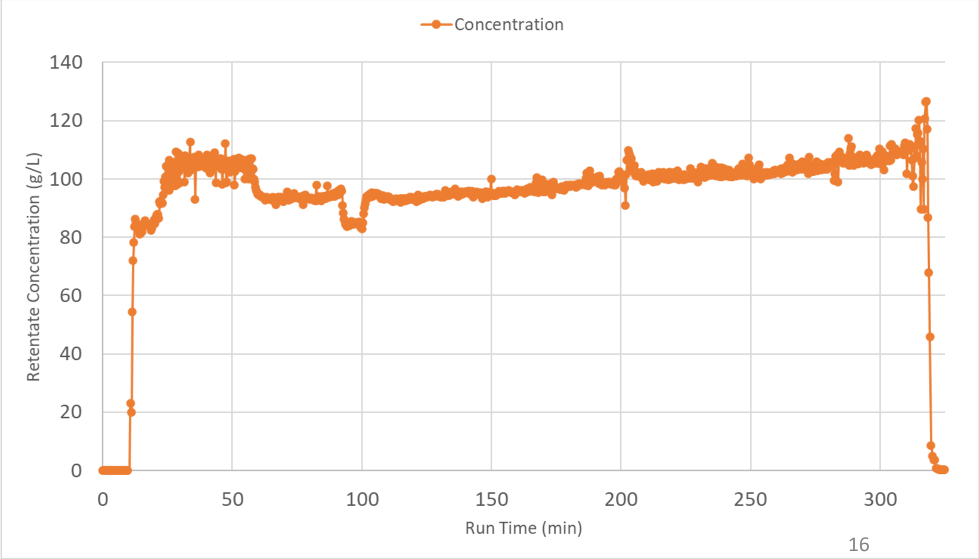
199-201 min – Feed pressure transmitter was broken.

302 min – Switched to buffer for flushing









Counter Current Buffer Exchange

Expected Results

- Buffer exchange
  - Crossflow 0.6 LMM (36 LMH)
  - Expected TMP 12 psi

Notes

Feed Conc: 100 g/L

Retentate Conc: 100 g/L

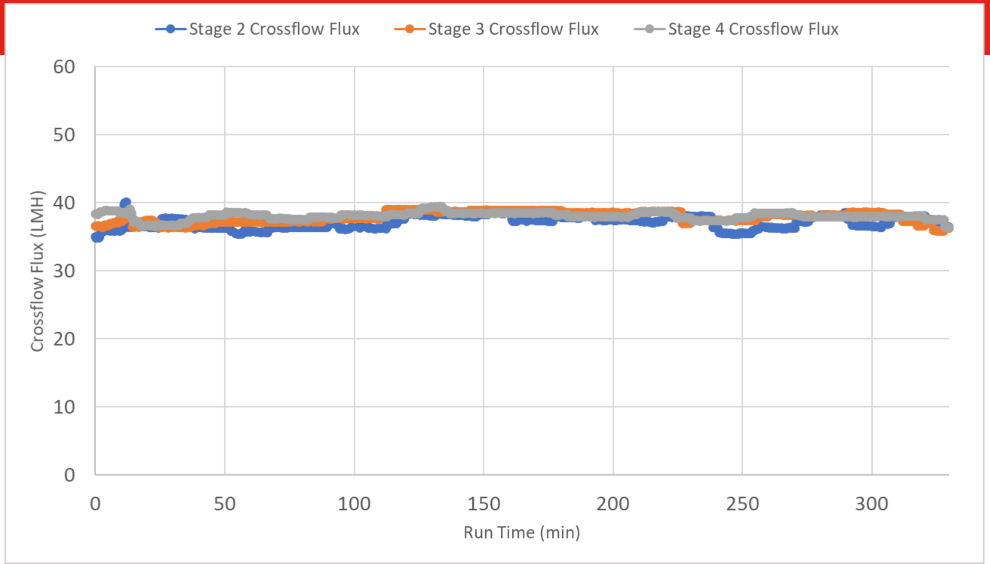
0 min – Began recording data

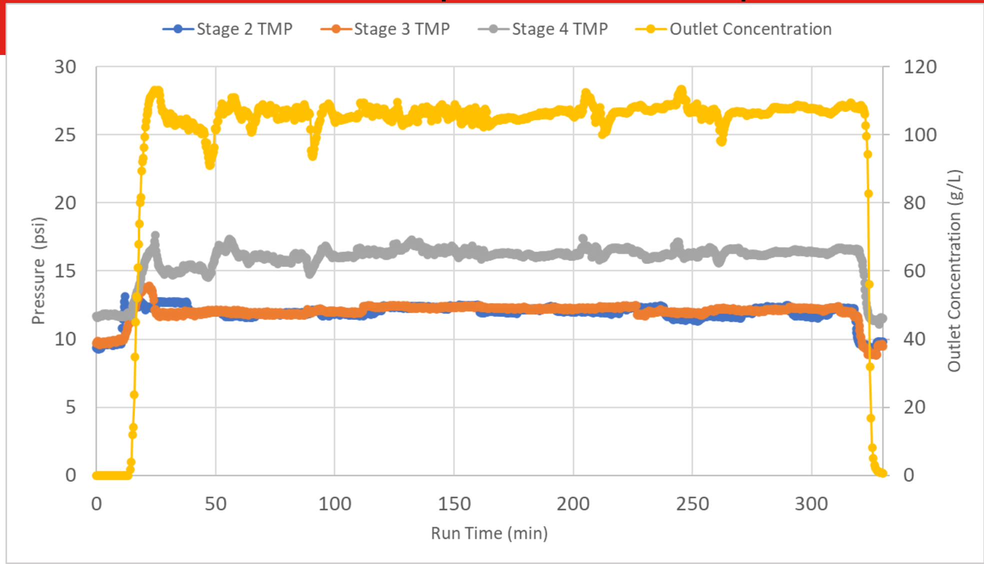
9 min – Started flowing feed at 36 LMH (144 mL/min)

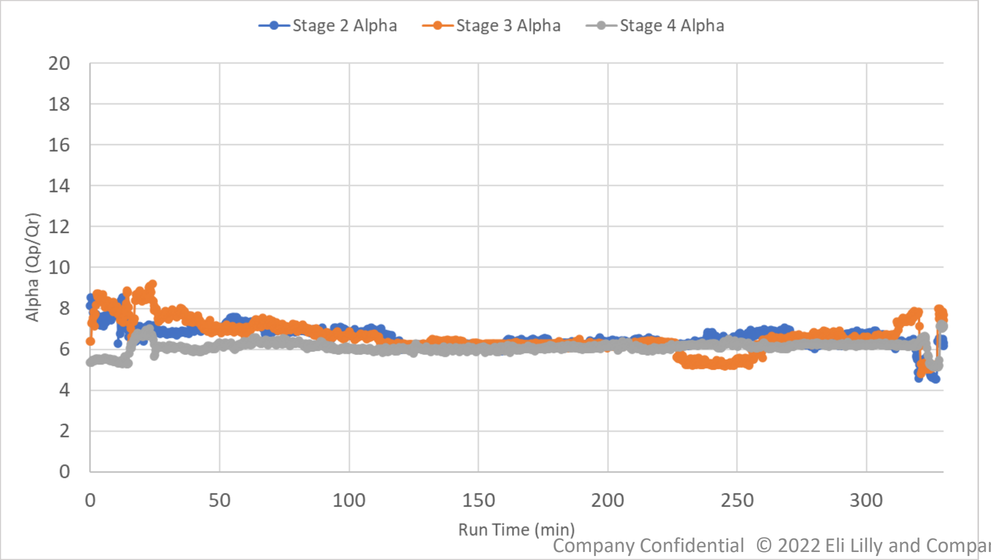
241 min – Switched to buffer flush

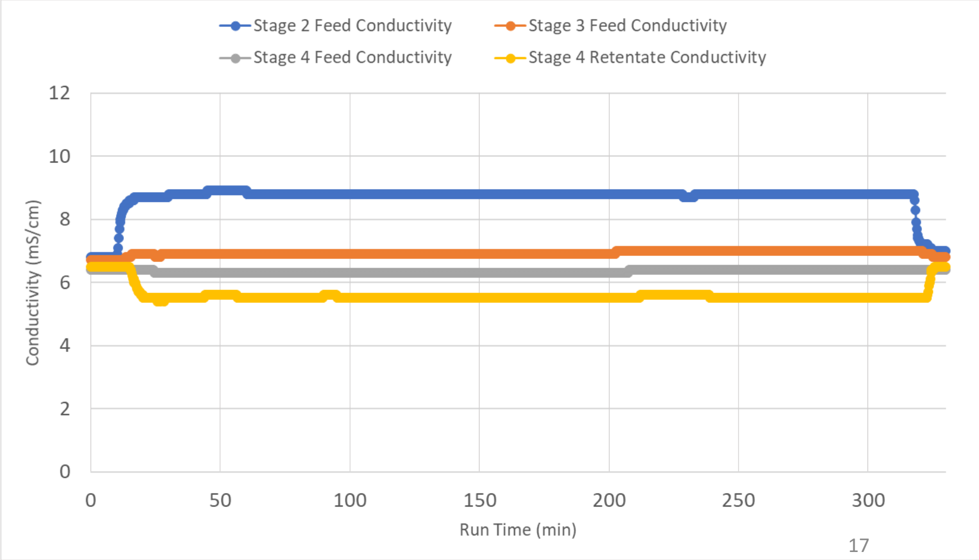
257 min – Started cleaning (16 min shutdown)

Adjusted pump speeds (feed and permeate) as needed to maintain tank levels









Secondary Concentration

Results

- Secondary concentration
  - Crossflow 0.3 LMM
  - Expected TMP 6 psi

Notes

Feed Conc: 100 g/L

Retentate Conc: 250 g/L

0 min – Began recording data

1 min – Started feed pump

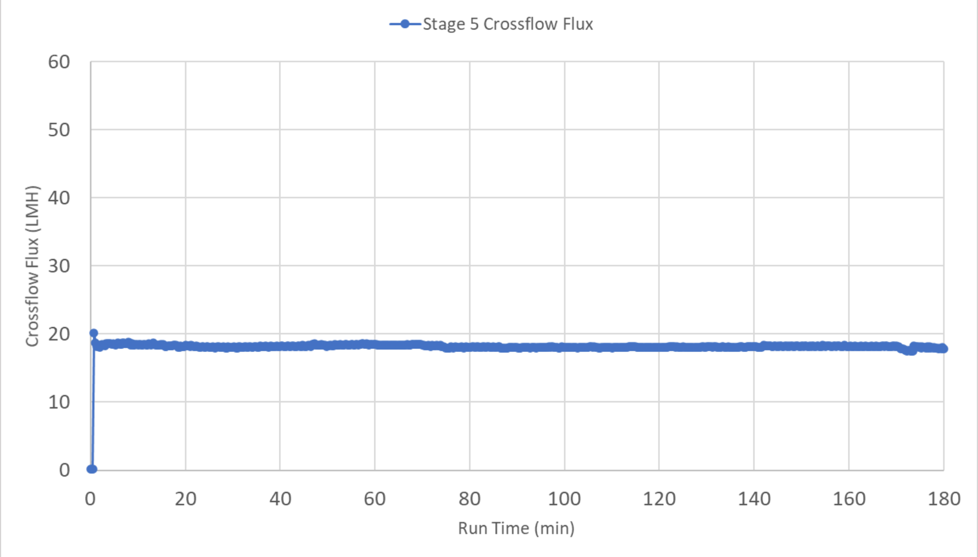
1-15 min – Adjusted retentate valve to target retentate conc of 200 g/L

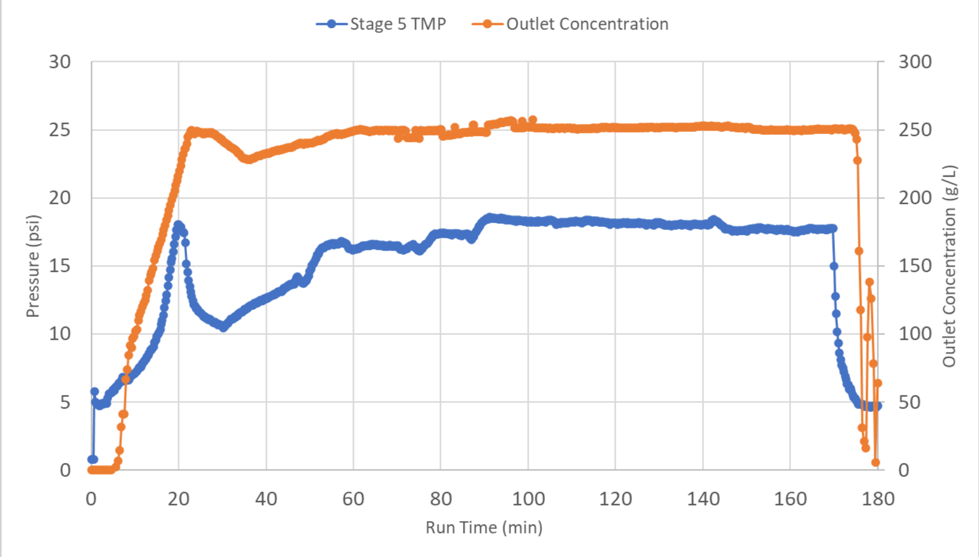
22 min – Adjusted retentate valve back to hold steady at 250 g/L

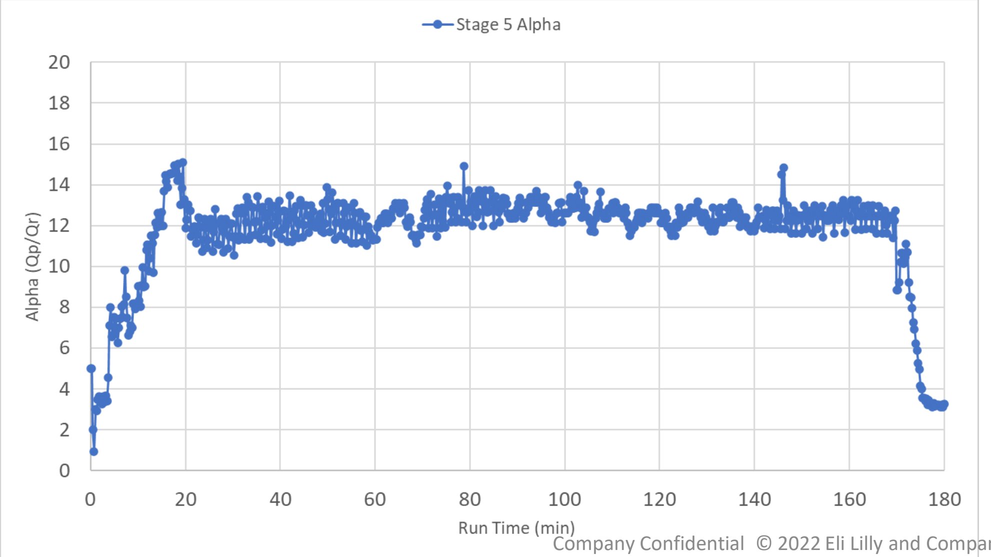
170 min – Started buffer flush

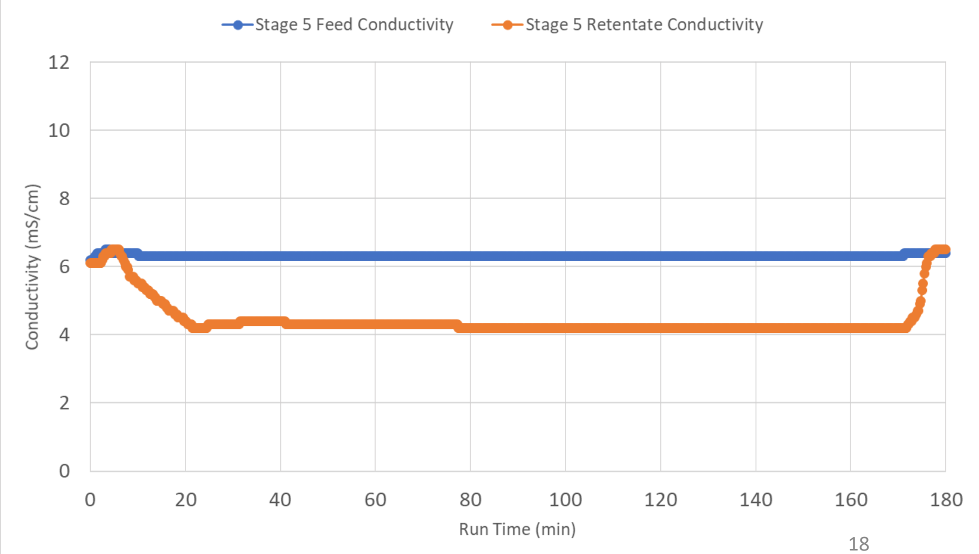
187 min – Started cleaning routine (17 min shutdown)

Adjusted pump speed as needed to maintain tank level









cTFF POC Results

Material	UV (g/L)	Viscosity (cP)	Density (g/mL)	pH	Conductivity (mS/cm)	Purity (SEC)
CEX Feed	11.5	1.0	1.0107	4.77	21.20	99.12
Exchange Buffer		1.0	1.0035	5.73	5.72	
Primary Conc. Retentate	106.8			5.67	5.06	98.95
CCBE Retentate	107.5	2.8	1.0342	5.63	7.53	98.93
Secondary Conc. Retentate	224.8	23.4	1.0702	5.75	3.84	98.68

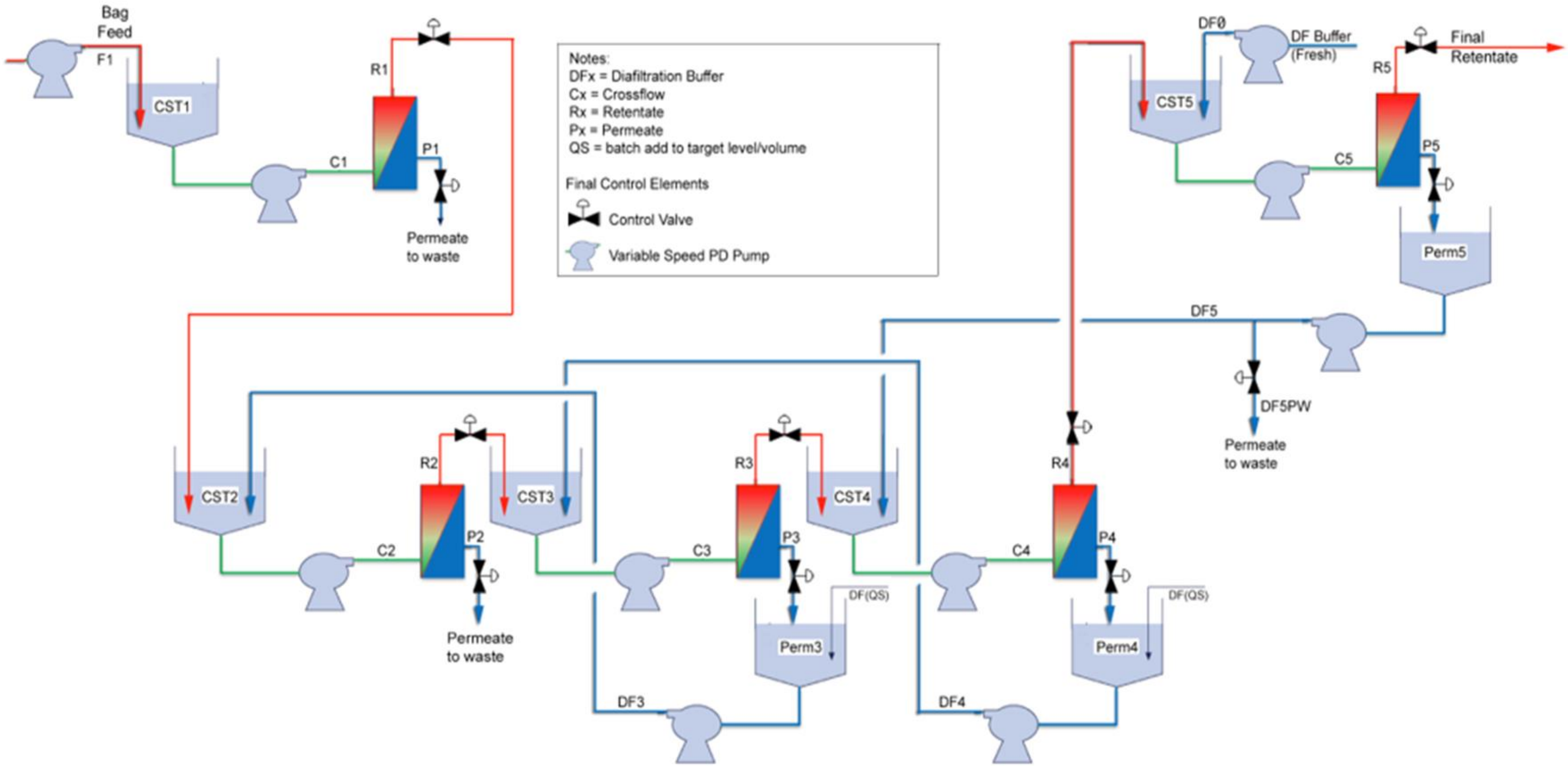
- Concentration Targets
  - Primary – 100 g/L
  - CCBE – 100 g/L
  - Secondary – Started at 200 g/L and increased to 250 g/L (average 225 g/L)
    - 10x increase in viscosity from 100 g/L to 225 g/L

Financial-Operational Analysis

bTFF vs. cTFF

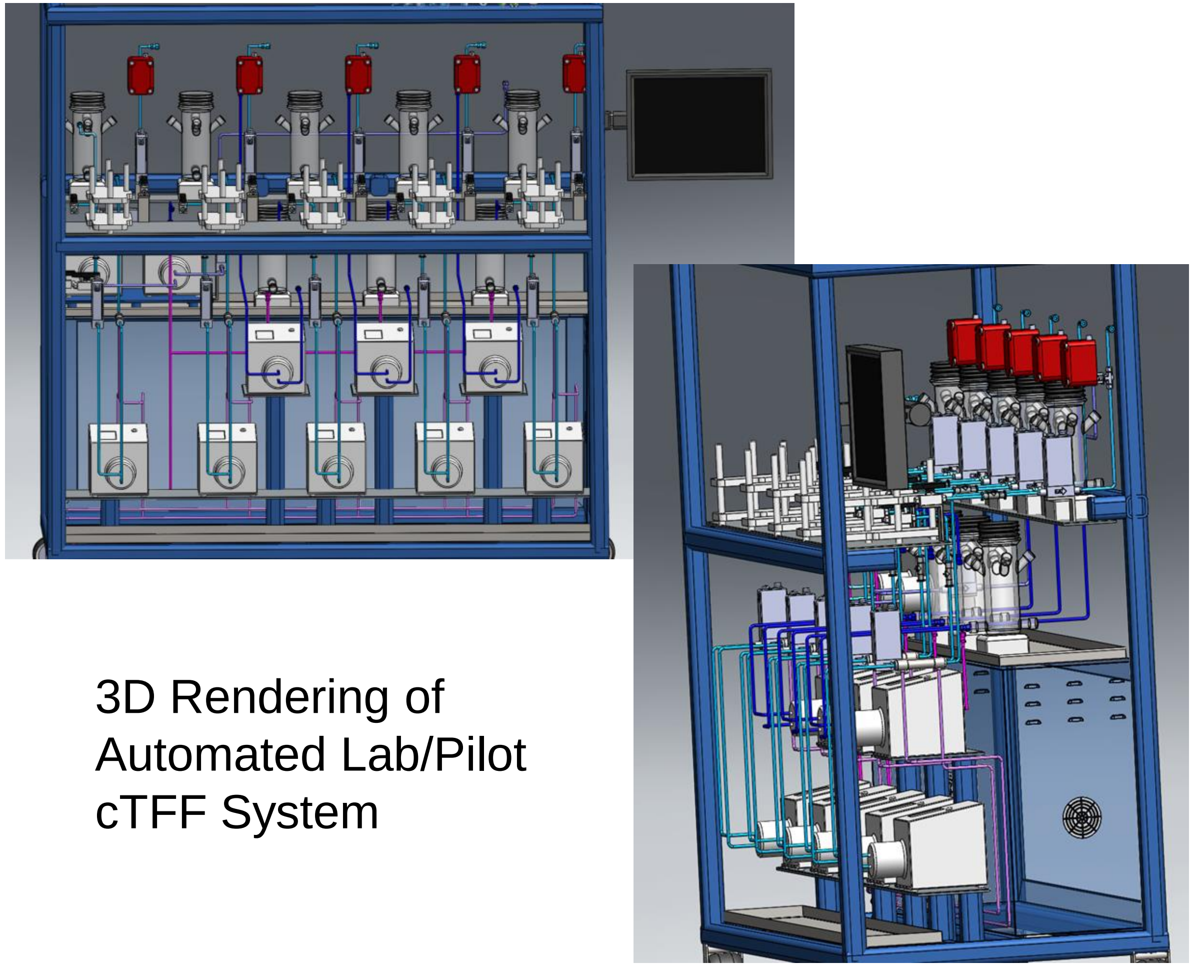
- Basis
  - High throughput mAb
    - Metric tons / year
- Negatives
  - Increase in surface area
- Positives
  - Reduction in single use mixers (SUMs)
  - Reduction in buffer consumption vs batch
  - Reduction in operator interaction time
- Annual cost savings

Version 4.0 Design



Wrap-Up Slide

- cTFF has been demonstrated using integrated SP-TFF and CCBE-TFF
- cTFF is a viable option for next generation continuous purification
- Lilly to execute nonGMP pilot scale cTFF with a fully automated system in 2023



3D Rendering of Automated Lab/Pilot cTFF System

